Denise Case, PhD, PE (Retired, Kansas)

Northwest Missouri State University, Associate Professor

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Professional Preparation

- PhD Computing & Information Science, Kansas State University, 2015
- Master of Software Engineering, Kansas State University, 2013
- BS Chemical Engineering, Honors, University of Missouri at Columbia, 1985

Academic Appointments

Northwest Missouri State University (2015-present)

- Tenured Associate Professor of Computer Science, Aug 2020-Aug 2022
- Assistant Professor of Computer Science, Aug 2015 - Aug 2020
- Lecturer, Online Data Analytics Program, Aug 2022 - Present; post-tenure appointment

Johnson County Community College (2010-2014)

- Adjunct Assistant Professor, Information Systems (2010-2014)- Instructor for concepts of programming algorithms with C# and introductory programming courses; developed OO course in C#, including LINQ, intro to complexity analysis, and simple AI projects.
- Curriculum Development - Developed the course materials for the Johnson County Community College Computer Science Object-Oriented Programming Using C# class. Assisted with the development of the course description and requirements.

Kansas State University (2011-2015)

- Graduate Research Assistant - Developed multigroup agent architecture for complex cyber-physical systems; used the architecture to simulate a complex CPS for an intelligent power distribution system to evaluate power quality control algorithms with high penetration of intermittent photovoltaic generation sources and evaluate game-theoretic online auction algorithms near sources of distributed generation.
- NSF INSIGHT Fellow - Collaborated with National Science Foundation, K-State, and Kansas Schools on the Infusing System Design and Sensor Technology in Education Fellowship to bring creative examples of the use of sensor technologies into GK-12 classrooms.

Research Program

Durable Skills in the Age of Generative AI (Education, Professional Practice)

- Research and develop practices focused on sustaining human judgment, accountability, and technical competence in AI-augmented environments.
- Investigate durable cognitive and professional skills essential in environments augmented by generative AI systems.
- Emphasize structured reasoning, reproducible workflows, and accountable use of AI tools in analytics and software development.
- Developed course materials and professional guides integrating version control, project structure, CI/CD, GitHub automation, professional documentation, and ethical AI use.
- Submitted letter to *ACM AI Letters* addressing durable learning competencies and capacity formation in the age of rapid AI capability expansion.

Data Analytics Education and Engagement

- Design online graduate analytics courses with strong student outcomes and evaluations.
- Develop industry-informed analytics curriculum emphasizing real-world skills, modeling decisions, and critical interpretation.
- Design and deliver project-based courses in professional foundations, continuous intelligence, streaming data, and business analytics.
- Integrate reproducible computing standards (Python project structure, version control, CI/CD) into analytics instruction.
- Foster student engagement through applied professional projects and scaffolded professional workflows.
- Create courses that engage those new to programming as well as senior executives catching up on modern best practices and innovations.

Reproducible and Explainable Infrastructure

- Design and develop generative AI concept exploration tools
- Implement QR code web application (GitHub Pages) for civic organizations
- Prepare targeted CI/CD and project file templates (Python, MkDocs, LaTeX, Lean)
- Publish Professional Python Guide (pro-analytics-02) using uv, pyproject.toml
- Publish Professional Python Guide (pro-analytics-01) using pip, venv, requirements.txt
- Publish PyPi package (datafun-toolkit) for privacy-aware professional Python logging

Structural Explainability (Founder and lead)

- Define characteristics of neutral substrates that enable coordination, identity persistence, and accountable change under persistent disagreement
- Define specifications: Accountable Entities (AE), Exchange Protocol (EP), Contextual Evidence and Explanation (CEE), and Governance and Interpretation Boundaries (GB, IB)
- Implement Lean formalizations of neutrality theorem and identity regimes

Civic Interconnect: Data Infrastructure (Founder and lead)

- Design and implement open-source infrastructure for structured data integration and interoperability.
- Develop identity-stable data architectures grounded in Structural Explainability.
- Build agent-based monitoring tools for schema evolution, jurisdiction mapping, and legislative data ingestion.
- Develop provenance-aware specifications and tooling for reproducible data workflows.
- Develop and maintain multi-repository open-source ecosystem supporting datasets, applications, and developer tooling.
- Validated identity stability and canonicalization logic using large-scale public procurement data (Chicago contracts dataset; 20,000-record test deployment).

Applied Ontology and Neutral Substrates

- Neutrality theorem (formal and narrative exposition)
- Identity regimes (six necessary and sufficient identity and persistence regimes)

Grants and Funded Research

- NSF Research Project under direction of A. Pahwa, (co-PIs) B. Natarajan, S. Das, S. A. DeLoach, X. Ou. Components, Run-time Substrates, and Systems: Medium: Holonic Multi-Agent Control of Intelligent Power Distribution Systems. National Science Foundation

(NSF), September 2011 - August 2015. $1,100,000. http://ipds.cis.ksu.edu/.
- NSF S-STEM Students (co-investigator, mentor). Advised Graduate Student on Agent-Based (AI) confidential entrepreneurial application involving Natural Language Processing, Sentiment Analysis, Classification, and Question Answering Capabilities.

Professional Experience

Black & Veatch (2004-2016)

- Senior Consulting Engineer - Consultant and analyst on the strategic solutions team for Smart Integrated Infrastructure ASSET360 Smart Analytics, an advanced online analytics platform for smart grid, smart water, and smart communities.
- Senior Client Solutions Manager - Lead analyst and system designer for $3M on-line energy efficiency and environmental compliance dashboard for utility clients; senior database designer and architecture consultant for enterprise monitoring and diagnostics application; reliability and executive information systems project manager; outstanding client satisfaction on technically challenging projects.

Case Consulting (1997-2004)

- Engineering & Information Systems Consultant - Created and managed successful consulting business for engineering and information systems design and implementation serving engineering, utility, regulatory, and marketing companies; developed new customers across diverse industries; key strategic role in innovative data marketing project; gained new and repeat business with high client satisfaction.

Black & Veatch (1987-1997)

- Senior Engineering Consultant, Environmental Division - Inter-division partnership to share experience across business sectors and provide environmental, permitting, and regulatory services to water and wastewater utility customers.
- Manager, BV Technologies, Environmental Compliance Division - Managed compliance division for new technology venture.
- Project Manager and Engineering Manager - Managed multiple design and implementation teams of 5 to 15 engineers for power and environmental clients.
- Supervisor, Strategic Information Systems, Power Division - Managed team of ten engineers designing and implementing information systems for improved operations, performance, maintenance, and environmental compliance for electric power companies and organizations.
- Chemical Engineer, Power Division - Developed engineering specifications for treatment systems.

Courses Taught

Northwest Missouri State University: Online Master of Data Analytics

- CSIS 44608: Data Analytics Fundamentals
- CSIS 44613: Data Visualization
- CSIS 44620: Web Mining and Applied Natural Language Processing
- CSIS 44630: Continuous Intelligence and Interactive Analytics
- CSIS 44632: Business Intelligence and Analytics
- CSIS 44661: Databases for Analytics
- CSIS 44670: Applied Machine Learning

- CSIS 44671: Streaming Data
- CSIS 44688: Data Analytics Capstone Project

Northwest Missouri State University: Master of Applied Computer Science

- NW 44-517 Big Data (2018-2022)
- NW 44-542 Object-Oriented Programming. (2015, 2021)
- NW 44-563 Developing Web Applications and Services. (Spring 2017, Fall 2016)
- NW 44-560 Advanced Topics in Database Systems. (Spring 2016, Fall 2015)
- NW 44-564 Design of Data Intensive Systems. (Spring 2017)
- NW 44-618 Project Management in Business and Technology. (Summer 2021, Summer 2017, Summer 2016)
- NW 44-663 Component-Based Software Engineering: Application Development in C#.NET. (Spring 2016)
- NW 44-691 Graduate Directed Project I. (Fall 2021)
- NW 44-692 Graduate Directed Project II. (Spring 2022, Fall 2016)
- NW 44-696 Graduate Directed Project. (Spring 2016, Fall 2015)

Northwest Missouri State University: Undergraduate Computer Science

- NW 44-386 Digital Forensics. (2021, 2020, 2019)
- NW 44-499 Independent Study Computer Science. (2017, Spring 2016, Fall 2016)

Johnson County Community College

- CS 236 - Object-Oriented Programming Using C# (Spring 2013, Spring 2012)
- CS 201 - Concepts of Programming Algorithms using C# (Spring 2012, Fall 2011, Spring 2011, Fall 2010)
- Computer Science, CS 134 - Programming Fundamentals. (Fall 2010, Summer 2010)

Courses Developed

Northwest Missouri State University

- CSIS 44608: Data Analytics Fundamentals
- CSIS 44630: Continuous Intelligence and Interactive Analytics
- CSIS 44632: Business Intelligence and Analytics
- CSIS 44671: Streaming Data
- NW 44-517 Big Data (new Spring 2018)
- NW 44-563 Developing Web Applications and Services (major update: Fall 2016)
- NW 44-564 Design of Data Intensive Systems. (new Spring 2017)
- NW 44-663 Component-Based Software Eng: App Development in C#.NET. (New Fall 2017)
- NW 44-663 Component-Based Software Eng: App Development in C#.NET. (New Spr 2016)
- NW 44-691 Graduate Directed Project I
- NW 44-692 Graduate Directed Project II (major update: 2016, with Dr. Charles Hoot)
- NW 44-386 Digital Forensics
- NW 44-499 Independent Study Computer Science.

Johnson County Community College

- Computer Science CS 236 - Object-Oriented Programming Using C# (New Spring 2012)
- Computer Science CS 201 - Concepts of Programming Algorithms using C# (New Fall 2010)
- Computer Science, CS 134 - Programming Fundamentals. (New Summer 2010)

Selected Publications

- Case, Denise M. "Substrate Stability Under Persistent Disagreement: Structural Constraints for Neutral Ontological Substrates." *arXiv preprint arXiv:2601.16152* (2026).
- Case, Denise M. "The Ontological Neutrality Theorem: Why Neutral Ontological Substrates Must Be Pre-Causal and Pre-Normative." *arXiv preprint arXiv:2601.14271* (2026).
- Bandi, A., Case, D., Eloe, N., Fellah, A., & Hoot, C. (2022). Teaching multiple graduate sections with large class sizes: panel discussion. *Journal of Computing Sci in Colleges*, *37*(6), 73-73.
- Bandi, A., Case, D., Eloe, N., Fellah, A., & Hoot, C. (2022). Challenges developing and teaching online professional courses for technical graduate programs: panel discussion. *Journal of Computing Sciences in Colleges*, *37*(6), 71-72.
- Badami, Charles, et al. "Engaging graduate students during the pandemic: panel discussion." *Journal of Computing Sciences in Colleges* 36.6 (2021): 78-79.
- Case, Denise M. "Hey Alexa! a fun introduction to AWS Lambda, serverless functions, and JavaScript with conversational AI: conference workshop." *Journal of Computing Sciences in Colleges* 36.6 (2021): 66-66.
- Case, Denise M., and Michael P. Rogers. "Is It Getting Foggy in Here? Cloud computing in the classroom." *Journal of Computing Sciences in Colleges* 35.6 (2020): 92-106.
- Case, Denise M., Colton Steeve, and Matthew Woolery. "Progressive Web Apps are a Game Changer! Use Active Learning to Engage Students and Convert Any Website into a Mobile-Installable, Offline-Capable, Interactive App." *Proceedings of the 51st ACM Technical Symposium on Computer Science Education*. 2020.
- Case, Denise M., and Charles Hoot. "Capstone as consulting." *Journal of Computing Sciences in Colleges* 34.4 (2019): 67-77.
- Shadinger, D., Katsion, J., Myllykangas, S., & Case, D. (2020). The impact of a positive, self-talk statement on public speaking anxiety. *College Teaching*, *68*(1), 5-11.
- Case, Denise M., Ty Blackburn, and Chrysostomos Stylios. "Construction Industry Pursuit Intel Modeling Insights with FCM." *2019 IEEE International Conference on Fuzzy Systems (FUZZ-IEEE)*. IEEE, 2019.
- Bandi, A., Case, D., Eloe, N., & Fellah, A. (2019). Challenges of mentoring graduate directed projects: profession-based learning through collaboration. *Journal of Computing Sciences in Colleges*, *34*(4), 123-127.
- Badami, Charles, et al. "Blended courses in computer science and information systems education: Adapting to changing educational methods and needs." *Journal of Computing Sciences in Colleges* 34.4 (2019): 122-122.
- Case, Denise M., Ty Blackburn, and Chrysostomos Stylios. "Modelling construction management problems with fuzzy cognitive maps." (2018).
- Case, Denise, and Douglas Hawley. "Chrome Home: Six Fun Activities Introducing Basic Web Programming Techniques." *Proceedings of the 49th ACM Technical Symposium on Computer Science Education*. 2018.
- Nayak, Prasunjit, et al. "Using fuzzy cognitive maps to model university desirability and selection." *2017 IEEE Intl Conference on Systems, Man, and Cybernetics (SMC)*. IEEE, 2017.
- Case, Denise M., and Charles Hoot. "Developing a lessons library for teaching version control." *Journal of Computing Sciences in Colleges*, 32.5 (2017): 88-94.
- Denise M. Case, Chrysostomos Stylios, “Introducing a Fuzzy Cognitive Map for Modeling Power Market Auction Behavior”. At 2016 IEEE Symposium Series on Computational Intelligence, Athens, Greece, December 8, 2016.
- Denise M Case, Nathan Eloe, Jennifer Leopold, “Scaffolding Version Control into the Computer Science Curriculum”. At 22nd International Conference on Distributed

Multimedia Systems, Salerno, Italy, November 26, 2016.

- Nathan Eloe, Denise M Case, Jennifer Leopold, "VeCVL: A Visual Language for Version Control". At 22nd International Conference on Distributed Multimedia Systems, Salerno, Italy, November 25, 2016.
- Denise M Case, Chrysostomos Stylios, "Fuzzy Cognitive Map to Model Project Management Problems". At 35th North American Fuzzy Information Processing Society Annual Conference, El Paso, TX, USA November 3, 2016.
- Diana Linville, Michael P. Rogers, Curt Kelly, Carol Spradling, Denise M. Case, "Profession-based learning through collaboration and vertical alignment with K12, higher education, and industry: panel discussion". In Journal of Computing Sciences in College, 2016, presented at the Consortium for Computing Sciences in Colleges Conference, May 2016.
- Denise M. Case, M. N. Faqiry, B. P. Majumder, S. Das, and Scott. A. DeLoach. "Implementation of a two-tier double auction for on-line power purchasing in the simulation of a distributed intelligent cyber-physical system". In Research in Computing Science, 2014, to be presented at the Mexican International Conference on Artificial Intelligence, 2014.
- Denise M. Case. "Distributed Intelligent Systems: Agent Self Control and Consistency Management in Complex Adaptive Systems," to be presented at the Mexican International Conference on Artificial Intelligence, Doctoral Consortium 2014.
- Denise M. Case and Scott. A. DeLoach. OBAA++: An Agent Architecture for Participating in Multiple Groups. In Proceedings of the Thirteenth International Conference on Autonomous Agents and Multiagent Systems, 2014, http://aamas2014.lip6.fr/.
- Denise M. Case, Engineering multigroup agents for complex cooperative systems, Doctoral Consortium, 13th International Conference on Autonomous Agents and Multiagent Systems, AAMAS 2014. http://aamas2014.lip6.fr.
- Denise M. Case and Scott A. DeLoach. Applying an O-MaSE Compliant Process to Develop a Holonic Multiagent System for the Evaluation of Intelligent Power Distribution Systems, 1st International Workshop on Engineering Multi-Agent Systems (EMAS) Workshop (2013) at the Twelfth International Conference on Autonomous Agents and Multiagent Systems, http://aamas2013.cs.umn.edu/.
- Anil Pahwa, Fellow, IEEE, Scott A. DeLoach, Member, IEEE, Bala Natarajan, Senior Member, IEEE, Sanjoy Das, Ahmad R. Malekpour, Student Member, IEEE, S M Shafiul Alam, Student Member, IEEE, and Denise M. Case, Member, IEEE, Goal-Based Holonic Multi-Agent System for Operation of Power Distribution Systems, IEEE Transactions on Smart Grid (Special Issue on Cyber Physical Systems and Security for Smart Grid), 2014.

Service (Selected)

University Service

- Faculty Senate, Secretary (2019-2020)
- MS-ACS Program, Curriculum, and Professional Advisory Committees (2015–2022)
- Capital Planning Committee (2017-2018)
- Faculty Search and Textbook Review Committees
- Grant proposals and research acquisition (S-STEM, Sphero educational outreach, Proxmox server, GitSubmit, ioSense, Azure Educator, Splunk)

Professional Service

- ACM Graduate Student Chapter Faculty Advisor (2016-2020)
- ACM-W Communications Committee (2016-2020)
- Math Olympiad Invited Speaker (2016)

- CCSC-CP: Research presentations and student mentorship (2016-2022)
- NCWIT Higher Ed Alliance

Community and Civic Service

- American Association of University Women, Ely Branch Co-Secretary (2024-present)
- Web Manager, Minnesota Area 35 District 8 (2026-present)
- Johnson County Community College CSIT Advisory Board (2011-2014)

Professional Development (Selected)

- Continuing education in standards and cybersecurity (NIST; Google Cybersecurity, 2023)
- IBM Quantum Computers Programming- 4 Week Asynchronous Workshop (Oct 2021)
- IBM Big Data and Cognitive Computing workshops (2019–2020)

Certifications and Associations

- Professional Engineer, Licensed in Kansas, 12575 (inactive)
- Association of Computing Machinery (ACM) Lifetime Member
- Institute of Electrical and Electronics Engineers (IEEE)
- American Association of University Women (AAUW)
- American Institute of Chemical Engineers, Kansas City Section, President
- Tau Beta Pi Engineering Honorary, Lifetime Member
- Power Masters Toastmasters Club, President
- Omicron Delta Kappa Leadership Honor Society, Secretary
- Alpha Chi Sigma Technical Fraternity, President